



School: Campus:
Academic Year: Subject Name: Subject Code:
Semester: Program: Branch: Specialization:
Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment : Build a Use Case – Tokenized Supply Chain

* Coding Phase: Pseudo Code / Flow Chart / Algorithm

1. **Identify Participants:**

Define all stakeholders — Manufacturer, Supplier, Transporter, Distributor, Retailer, and Customer.

2. **Token Creation:**

Create digital tokens representing goods or product batches. Each token contains product ID, origin, and value.

3. **Smart Contract Setup:**

Establish smart contracts that govern token transfer, payment release, and verification of product conditions.

4. **Product Registration:**

The manufacturer registers the product by assigning tokens that represent the items being shipped.

5. **Token Transfer Through Supply Chain:**

As the product moves through different stages, ownership of tokens is transferred from one participant to the next (e.g., Manufacturer → Supplier → Distributor → Retailer → Customer).

6. **Verification and Validation:**

Each transfer is verified by the blockchain network to ensure authenticity and prevent double- spending of tokens.

Software used:

1. VS Code.
2. MS Word.
3. Brave for researching.

*** Implementation Phase: Final Output (no error)**

- All participants in the supply chain are enrolled as authorized members of the blockchain network.
- The manufacturer generates digital tokens that symbolize the physical goods ready for shipment.
- Upon confirmation of material receipt, the tokens are securely transferred to the supplier.
- As the goods progress through the logistics chain, token ownership is sequentially handed over to the transporter, distributor, and retailer.
- Every transfer and transaction is permanently logged and verified on the blockchain ledger.
- Smart contracts automatically release payments once predefined delivery conditions are satisfied.
- The end consumer obtains both the physical product and its corresponding digital proof of ownership (token).
- The blockchain ledger maintains a transparent, tamper-proof record of the entire product journey from manufacturer to customer.

*** Observations:**

- Key participants like Manufacturer, Supplier, Distributor, Retailer, and Customer were defined in the blockchain network.
- Digital tokens were created to represent products, ensuring traceability and authenticity.
- Smart contracts automated token transfer, verification, and payment settlement across the supply chain.
- Each transaction was validated by the blockchain, preventing fraud and ensuring data integrity.
- All records were securely stored on the blockchain, providing transparency and trust throughout the process.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student:

Name :

Signature of the Faculty:

Regn. No. :